

Effects of Matching Short Current Density for Dual Junction Lead Salts TPV Device at 1400 C°

Majed Khodr

Electrical and Electronics Department
American University of Ras Alkhaimah
Ras Al Khaimah, UAE
majed.khodr@aurak.ac.ae

Patrick McCann

Electrical and Computer Engineering
University of Oklahoma
Norman, OK, USA
pmccann@ou.edu

Abstract—Thermophotovoltaic (TPV) device that can be efficiently used to convert thermally stored energy in molten silicon at 1400 C° cells to electrical energy has been studied. The device utilizes $\text{Pb}_{0.8}\text{Sr}_{0.2}\text{Se}$ / $\text{Pb}_{0.81}\text{Sn}_{0.19}\text{Se}$ / $\text{Pb}_{0.8}\text{Sr}_{0.2}\text{Se}$ IV-VI material in a multiple quantum well configuration two junction device structure. Using quantum well calculations and including strain effects the interband optical transition energy gaps (well widths) were calculated to be 450 meV (3.5 nm), and 343 meV (4.9 nm), for the top and bottom junctions, respectively. We calculated the short current density, output electric power density and efficiency for each junction as a function of device thickness. The overall maximum electric power density and efficiency for the device was 3.46 W/cm² and 26.7% at device thickness of 6.0 μm ; however the short current density for both junctions was not matched. In order to do this and stabilize the device performance, we optimized the junction layers n and p thickness values. As a result, the device maximum output electrical power and efficiency dropped to 3.05 W/cm² and 16.89 % at device thickness values of 3.65 and 9.95 μm . At these values, the matched short circuit current density is 11.55 A/cm². It is shown that multiple quantum well single junction design for the above material system suffice to obtain better efficiency and output electric power compared to designs using the same bulk material system.

Keywords—thermophotovoltaic, short circuit current, efficiency, lead salts, MQW

I. INTRODUCTION

Thermophotovoltaic (TPV) devices are semiconductor diodes that convert photons from a blackbody radiating source into electricity. Recently extensive research emerged to utilize these devices in several applications such as combined heat and power [1,2], portable energy sources [3], space power [4], and energy storage systems [5,6]. The big advantage of TPV is that the emissivity of the source can be tailored to match the optical band gap of the semiconducting photo diode so it is possible to achieve good optical-to-electrical conversion efficiency. Low band gap materials for TPV devices have potentially important applications in the field of direct energy generation. A number of semiconductor compounds have very low bandgaps, e.g. InSb (0.17 eV), InAs (0.35 eV), InGaAsSb (0.3–0.7 eV) and PbS (0.37 eV). However, the strong non-radiative recombination mechanisms existing in those low bandgap materials especially Auger should be taken into account. However, semiconductor nanostructures such as quantum wells have the ability of suppressing Auger recombination [7]; thus, they must be considered to be of particular interest for low bandgap TPV cell developments. Hence, low band gap IV-VI semiconductor MQW materials are attractive for multi-junction TPV cell fabrication.

The bandgaps of $\text{Pb}_{0.8}\text{Sr}_{0.2}\text{Se}$ ($E_g = 840$ meV) / $\text{Pb}_{0.81}\text{Sn}_{0.19}\text{Se}$ ($E_g = 84$ meV) MQW materials alloys are smaller than those that can be reliably obtained with III-V semiconductor alloys. This significantly increases the amount of mid-IR emission that can be absorbed from a thermal radiator. The dual junction TPV cell made from these two alloys was analyzed at 1215 C°, optimized, and compared to the same bulk material [2, 6]. The critical thermal temperature of interest in this work is 1400 C°, as would likely be the case for latent heat energy storage in molten silicon. Figure 1 is a drawing of the dual junction TPV device structure where the bandgaps and subband energies in the top and bottom wells are shown. We defined 1.1 eV as an upper limit on the incident photon energy, and the two spectral region for absorption by the top and bottom junctions with band gaps of 0.450 eV and 0.343 eV, respectively. As noted in previous publication, the lattice parameter mismatch between the thin layers of $\text{Pb}_{0.8}\text{Sr}_{0.2}\text{Se}$ ($a_o = 0.615$ nm) and $\text{Pb}_{0.81}\text{Sn}_{0.19}\text{Se}$ ($a_o = 0.610$ nm) in the MQW material will create 0.41% compressive and tensile strains, respectively, in each layer [2, 6]. This strain, however, will slightly alter the well and barrier bandgaps through deformation potential effects [8]. In-plane compressive strain of the $\text{Pb}_{0.8}\text{Sr}_{0.2}\text{Se}$ barrier layer will increase its bandgap, while in-plane tensile strain of the $\text{Pb}_{0.81}\text{Sn}_{0.19}\text{Se}$ well layer will decrease its bandgap thus making well depths deeper. As a result, the newly calculated subband gaps for the top and bottom junctions will be 0.443 eV and 0.337 eV, respectively. In this work, we optimized the top and bottom junctions' thicknesses in calculating the TPV device efficiency and output electric power. The analysis was done at 1400 C°, using the model and material parameters shown in Table I, Fig. 1, and published previously [2, 6]. The dual junction TPV device modeled here can serve as a benchmark design for development of future TPV devices with high power conversion efficiencies.

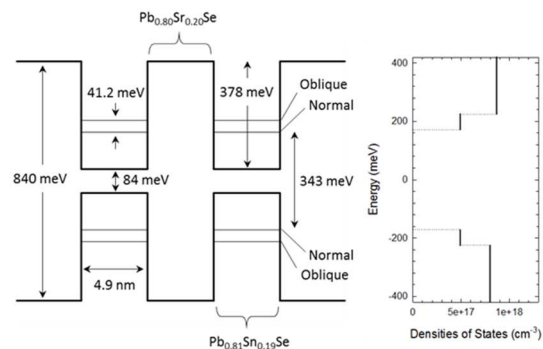


Fig. 1. Top and bottom junctions subband energies and corresponding densities of states [2].

TABLE I. MODEL PARAMETERS

Model Parameters	Values
Operating Temperature(K)	300
Carrier life time (ns)	10
Source emissivity	0.9
Min radiation spectrum wavelength (μm)	0
Max radiation spectrum wavelength (μm)	30
Max hot photons radiation band wavelength (μm)	1.13
Spectrum power radiance (w/m ² /sr)	127308
Spectrum power radiant emittance (w/m ²)	399950
Spectrum photon radiance (photons/s/m ² /sr)	2.04E+24
Spectrum photon radiant emittance (photons/s/m ²)	6.41E+24
Filtered hot band power radiance (w/m ² /sr)	6472
Filtered hot band photon flux radiance (photons/s/m ² /sr)	3.16E+22
Top layer electron mobility (cm ² /v.s)	800
Top layer hole mobility (cm ² /v.s)	800
Bottom layer electron mobility (cm ² /v.s)	800
Bottom layer hole mobility (cm ² /v.s)	800
Top layer holes doping density (cm ⁻³)	3x10 ¹⁸
Top layer electrons doping density (cm ⁻³)	3x10 ¹⁷
Bottom layer holes doping density (cm ⁻³)	3x10 ¹⁸
Bottom layer electrons doping density (cm ⁻³)	3x10 ¹⁷
Top layer absorption coefficient (1/m)	1.3x10 ⁶
Bottom layer absorption coefficient (1/m)	0.82x10 ⁶

II. METHODOLOGY

Thermophotovoltaic (TPV) cells' performance can be analyzed based on the output electrical power density and overall efficiency. The overall efficiency is given by the following relationship:

$$\eta = \frac{P_{out}}{P_{in}} = \frac{V_{oc} * J_{sc} * FF}{P_{in}} \quad (1)$$

where V_{oc} is the open circuit voltage, J_{sc} is the short circuit current density (A/cm²), FF is the filling factor, and P_{in} is the input optical power density (W/cm²).

The open voltage calculations can be done using the following formula:

$$V_{oc} = \frac{nkT}{q} * \ln\left(\frac{J_{sc}}{J_0} + 1\right) \quad (2)$$

where k is Boltzmann constant, T is device temperature in Kelvin, and n ideality factor assumed 1 in this work, J_0 is the dark current density in (A/cm²).

The short circuit current density calculations can be calculated using the following:

$$J_{sc} = q \cdot (P_n + N_p) \quad (3)$$

where P_n and N_p are the minority carrier reaching the end of junction as a result of the creation of hole-electron pairs in the n-type side and p-type side of the junction, respectively.

The shunt and series resistances are ignored in this work, and hence the filling factor depends only on the normalized

open circuit voltage $v_{oc} = \frac{V_{oc}}{nkT/q}$, and can be calculated using the following [9]:

$$FF = \frac{v_m}{v_m + 1} \frac{v_{oc} - \ln(v_m + 1)}{v_{oc} [1 - e^{-v_{oc}}]} \quad (4)$$

where $v_m = v_{oc} - \ln(v_m + 1)$ is the normalized open circuit voltage at the maximum power point of the cell.

Finally, inserting (2)-(4) in (1), we can calculate the overall efficiency as:

$$\eta_{TPV} = \frac{V_{oc} T * J_{sc} T * FF T + V_{oc} B * J_{sc} B * FF B}{P_{in, TPV}} \quad (5)$$

III. DATA AND RESULTS

The maximum obtained values for the short circuit current density, output electrical power density and efficiencies are shown in Table II. It is noticed that these maximum values occur at around total junction thickness value of 6.0 μm. This is due to the diffusion lengths of electrons and holes in relationship to the junction thicknesses of the n-type and p-type layers that comprise a dual junction TPV device. We assumed 60% and 80% absorption fraction for the top and bottom junction, respectively. This can be achieved because the junctions have more narrow absorption bands allowing specific optical coating thicknesses to be more effective across the band. Moreover, we used back surface reflection (BSR) of 0% and left this calculation for another publication.

Fig. 2 shows the calculated efficiency values for the dual junction device as a function of device thickness (solid line) and the efficiency for each junction as a function of junction thickness. However, these efficiency values are

TABLE II: MAXIMUM OUTPUT ELECTRICAL DENSITY AND EFFICIENCIES FOR EACH JUNCTION

Junction	Wavelength Range (μm)	Eg Range (meV)	Pin (W/cm^2)	Jsc (A/cm^2)	Pout (w/cm^2)	Maximum Efficiency (%)
Top	1-2.8	1240-443= 797	12.97	19.063	3.464	26.70
Bottom	2.8-3.68	443-337=106	5.09	11.551	1.086	21.32
Device	1-3.68	1240-337=903	18.069	unmatched	4.550	25.18

calculated before matching the short current density Jsc for both junctions, which can be seen in Fig. 3. Matching Jsc is important for the stability of the device, therefore we optimized the junction layers n and p thickness values. From Fig. 3, one can notice that the correct approach is to lower the Jsc values for the top layer by iterating the thickness values for the n and p layers to obtain a matched Jsc between the two junctions. The filled two circles in Fig. 2 shows two thickness values at which Jsc for the top and bottom junctions are matched. These two thickness values are $3.65 \mu\text{m}$ and $9.95 \mu\text{m}$. Table III shows the matched short circuit current density value $11.5 \text{ A}/\text{cm}^2$ for each junction, the output electrical powers, and maximum efficiency values for each junction and the overall device.

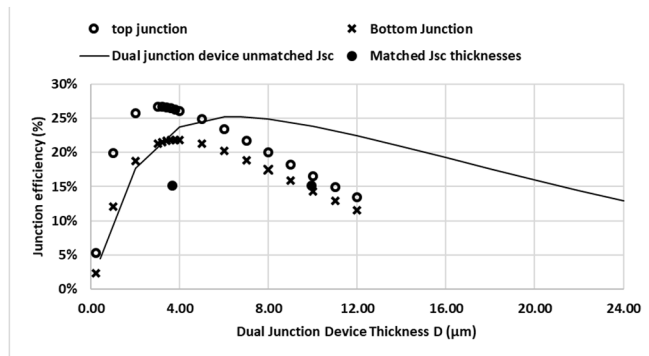


Fig. 2. Dual junction device efficiencies as a function of total device thickness (solid line). Top (open circle) and bottom (crossed) junctions' efficiencies as a function of their thicknesses. The filled circles present the thicknesses that correspond to the short current density matched values.

TABLE III: MAXIMUM OUTPUT ELECTRICAL DENSITY AND EFFICIENCY FOR EACH JUNCTION AFTER MATCHING J_{sc}

Junction	Pin (W/cm^2)	Jsc (A/cm^2)	Pout (w/cm^2)	Maximum Efficiency (%)
Top	12.975	11.537	1.964	15.14
Bottom	5.09	11.551	1.086	21.32
Overall	18.069	11.5	3.050	16.89

The maximum output electrical power and overall efficiency values are lower due to the matching process of the top and bottom junction, which resulted in lowering the short circuit current density from its high value for the top junction. Using the top junction only, i.e single junction, eliminates the need for this process, kept the output electrical power at $3.464 \text{ W}/\text{cm}^2$, and the efficiency value of 26.70% at a device thickness value of $3.0 \mu\text{m}$, as can be seen in Table I.

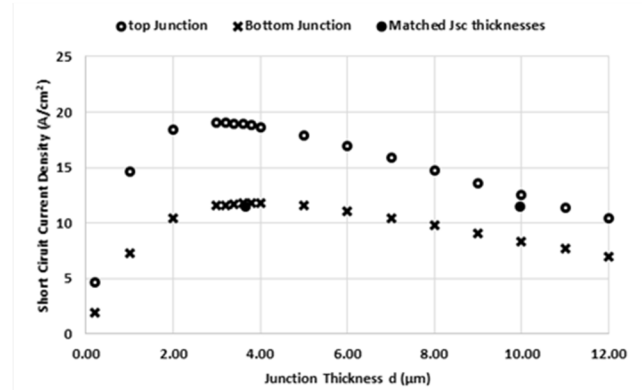


Fig. 3. The unmatched top and bottom junctions short-circuit current densities.

IV. CONCLUSION

The lead salts material systems analyzed in this paper can be attractive for the development of TPV-based devices. The obtained theoretical maximum output electrical density value of $3.05 \text{ W}/\text{cm}^2$ and overall efficiency value of 16.89% were obtained for 2J device design. However, improved values for the output electrical power and efficiency can be obtained at $3 \mu\text{m}$ device thickness for a single junction device. The findings from 1J is left for future publications.

REFERENCES

- [1] M. Bianchi, C. Ferrari, F. Melino, A. Peretto, "Feasibility study of a thermophoto-voltaic system for CHP application in residential buildings," *Appl. Energy*, vol. 97, pp.704-713, September 2012.
- [2] M. Khodr, M. Chakraborty, and P. J. McCann, "PbSnSe/PbSrSe Quantum Well Materials for Thermophotovoltaic Devices," *AIP Advances*, vol.9, pp. 035303-1- 035303, March 2019.
- [3] J W.R. Chan, et al., "Toward highenergy-density, high-efficiency, and moderate-temperature chip-scale thermophotovoltaics," *Proc. Nat. Acad. Sci. U.S.A.*, vol. 110 (14), pp. 5309-5314, April 2013.
- [4] M.R. Gilpin, D.B. Scharfe, M.P. Young, "Phase-change thermal energy storage and conversion: development and analysis for solar thermal propulsion," 48th AIAA/ASME/SAE/ASEE Joint Propulsion Conference & Exhibit, Atlanta, GE (US), 2012.
- [5] A. Datas, D.L. Chubb, A. Veeragavan, "Steady state analysis of a storage integrated solar thermophotovoltaic (SISTPV) system," *Sol. Energy*, vol. 96, pp. 33-45, October 2013.
- [6] M. Khodr, M. Chakraborty, P. J. McCann, "An innovative TPV device design based on narrow gap IV-VI semiconductor MQW structures," *Proc. SPIE 11121, New Concepts in Solar and Thermal Radiation Conversion II*, vol. 1112104, September 2019.
- [7] G.E. Cragg, A.L. Efros, "Suppression of Auger processes in confined structures," *Nano Lett.*, vol. 10, pp. 313-317, January 2010.
- [8] I. I. Zasavitskii, E. A. de Andrada e Silva, E. Abramof, and P. J. McCann, "Optical Deformation Potentials for PbSe and PbTe", *Phys. Rev. B*, vol. 70, pp. 115302-115312, September 2004.
- [9] M. A. Green, "Accuracy of analytical expressions for solar cell fill factors", *Solar Cells*, vol. 7, pp. 337-340, December 1982.